

Course: CSE407 Green Computing

Section 02

Term Project Report

Multi-Objective Optimization of Vehicle Speed for Simultaneous Energy Conservation and Travel Time Reduction using LSTM and NSGA-2

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Chapter 1

Introduction

Transportation systems account for a substantial share of global energy consumption and greenhouse gas emissions, making vehicle energy efficiency a critical research priority in the context of climate change, urban congestion, and sustainable mobility. While advances in vehicle electrification and powertrain technology have reduced tailpipe emissions, driving behavior remains a dominant factor influencing real-world energy consumption. Frequent acceleration, inefficient speed control, and poor anticipation of road conditions can significantly degrade vehicle efficiency, even in modern electric and hybrid vehicles.

Eco-driving strategies aim to address this challenge by optimizing vehicle speed and acceleration profiles to reduce energy consumption while maintaining acceptable travel time. Traditional eco-driving approaches often rely on rule-based heuristics or physics-based vehicle models. Although effective in controlled settings, these methods face limitations in scalability, adaptability, and computational cost when applied to complex, real-world driving conditions. High-fidelity physics-based simulations require detailed vehicle parameters and extensive calibration, while rule-based systems struggle to generalize across routes, traffic conditions, and vehicle types.

Recent advances in data-driven modeling and artificial intelligence offer promising alternatives. Large-scale vehicle datasets combined with deep learning techniques enable models to learn complex, nonlinear relationships between driving behavior and energy consumption directly from real-world data. In particular, recurrent neural networks such as Long Short-Term Memory (LSTM) models are well suited for capturing temporal dependencies in vehicle dynamics. When deployed as surrogate models, these data-driven predictors can replace computationally expensive simulations, enabling efficient optimization over long driving horizons.

At the same time, vehicle energy optimization is inherently a multi-objective problem. Minimizing energy consumption often conflicts with minimizing travel time, requiring explicit trade-off analysis rather than single-objective optimization. Multi-objective evolutionary algorithms, especially the Non-dominated Sorting Genetic Algorithm II (NSGA-II), have demonstrated strong

performance in solving such problems by generating a diverse set of Pareto-optimal solutions. However, many existing studies apply these techniques using simulated environments or simplified models, limiting their applicability to real-world driving scenarios.

Motivated by these gaps, this research proposes a fully data-driven framework that integrates deep learning-based surrogate modeling with multi-objective optimization to improve vehicle energy efficiency under realistic driving conditions. Using the Extended Vehicle Energy Dataset (eVED), which contains large-scale, real-world vehicle trajectory and energy data, an LSTM-based model is trained to predict short-horizon energy consumption from speed, acceleration, and road gradient. This model is then embedded within an NSGA-II optimization framework to generate eco-driving speed profiles that balance energy consumption and travel time for a fixed route.

Chapter 2

Literature Review

Recent literature on intelligent vehicle control, energy management, and routing optimization emphasizes the integration of advanced algorithms like NSGA-II and deep learning models to navigate complex, multi-objective constraints within dynamic transportation environments. The field grapples primarily with striking a balance between computational tractability, environmental sustainability, and passenger experience.

Hamednia et al. [11] concentrated on developing a computationally efficient algorithm for eco-driving over exceptionally long prediction horizons. The methodology framed the eco-driving scenario as a bi-level program: the bottom level optimized gear selection offline, while the top level solved a nonlinear dynamic program online, optimizing longitudinal velocity, kinetic energy, and acceleration using the real-time iteration sequential quadratic programming (RTI SQP) scheme within a Model Predictive Control (MPC) framework. This rigorous approach demonstrated significant utility by achieving energy savings of up to 15.71% compared to standard cruise

control. A primary obstacle addressed by this methodology is the pervasive "curse of dimensionality" inherent in traditional Dynamic Programming (DP) methods, where computational time escalates exponentially with the number of variables. However, an intrinsic limitation of methods utilizing Pontryagin's Maximum Principle (PMP), which informs their approach, is that they forfeit their computational advantage when state variables frequently encounter and activate their set bounds.

Similarly, El-Iali et al. [11] developed a robust multi-objective real-time energy management system (RT-EMS) tailored for Autonomous Plug-in Fuel Cell Electric Vehicles (APFCEVs), incorporating a hybrid system of a battery, fuel cell, and supercapacitor [18]. Utilizing a moving horizon approach and Quadratic Programming (QP), their model minimized hydrogen and electricity costs alongside fuel cell and battery degradation. The core utility lies in coupling the speed control system (a simplified MPC-based Adaptive Cruise Control [ACC]) with the lower-layer RT-EMS to provide highly efficient power distribution. This yielded results showcasing up to 99% optimality when compared against alternative State of Charge (SoC) planners. However, the study intentionally sidestepped real-time factors like route modifications or traffic variations, noting that including such complex, unpredictable dynamics would necessitate significantly higher computational resources.

Thematic Focus: Multi-Objective Optimization (MOO) via NSGA-II

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) has emerged as a predominant tool for balancing contradictory objectives in vehicle dynamics and routing problems.

Li et al. [23] designed an energy-efficient driving model by fusing Deep Convolutional Neural Network (DCNN) feature extraction with an enhanced NSGA-II algorithm and MPC. The fundamental utility of their work was achieved by enhancing the NSGA-II core with an adaptive crossover probability and elite retention mechanism, thereby addressing common shortcomings like sluggish convergence and ensuring the diversity of optimal solutions across scenarios. This optimization process resulted in dramatic improvements in urban congestion environments, specifically an increase of 25.42% in average vehicle speed and an 11.37% reduction in average traction power consumption.

Similarly, Lin et al. [4] applied NSGA-II to optimize acceleration duration, energy consumption, and jerk (driving comfort) for dual-motor EVs. Their comparative analysis demonstrated that NSGA-II generally exhibited a superior capacity for identifying solutions across the Pareto set when benchmarked against MOPSO, PESA-II, and SPEA2. The result was substantial: optimization yielded energy consumption savings of 37.44%. However, a place for improvement remains, as empirical results indicated that MOPSO occasionally demonstrated superior optimization capability under specific operating conditions, suggesting that refinement of crossover or mutation parameters could further enhance performance across different multi-objective algorithms.

However, Vasebi and Hayeri [15] identified a key systemic drawback of existing self-centered optimization: algorithms prioritizing only individual vehicle efficiency can paradoxically generate traffic congestion, thereby increasing network-level fuel consumption. To resolve this, they proposed Collective-Adaptive Cruise Control (Collective-ACC), using NSGA-II to formulate a bi-objective solution balancing direct fuel minimization with indirect traffic flow improvement. The utility of this approach was immediately apparent, yielding individual fuel reductions of up to 49% and lowering network-level CO₂ emissions by up to 54% in simulation. A clear limitation in their study, which presents a path for future research, is the estimation of production costs; the complexity model used to estimate the final cost of Collective-ACC excluded practical economic factors like R&D costs and marketing expenses, which are essential for automaker adoption decisions.

In the context of civic infrastructure, Alharbi [17] integrated neural forecasting of EV adoption with NSGA-II for sustainable infrastructure planning, simultaneously minimizing infrastructure costs, energy gap (demand versus available PV supply), and traffic congestion (travel distance to charging stations). The NSGA-II framework offered notable cost-efficiency, resulting in approximately 12.7% reduction in infrastructure costs and a 9.3% minimization of the predicted energy gap. The drawback identified was that while cost and energy estimations were reliable, predictive models for traffic congestion derived from adoption data showed considerably higher variability, underscoring the intrinsic complexity of modeling traffic dynamics.

Gülmez et al. [30] further leveraged NSGA-II in solving the Green Vehicle Routing Problem with Flexible Time Windows, targeting the minimization of economic cost, minimization of fossil fuel use, and maximization of customer satisfaction. The major finding confirmed the utility of adopting multiple flexible windows, which consistently led to reduced economic costs and higher customer satisfaction levels compared to using just one time window. This research, however, encountered the typical computational constraints of an NP-hard problem, implying that accurate, exact solutions remain constrained to small instances.

Li et al. [6] directed NSGA-II toward optimizing commuter carpooling under time-varying networks, establishing an Improved NSGA-II (INSGA-II) that merged CNN/LSTM temporal features with an open-loop saving algorithm and a 2-opt search strategy. This synthesis proved highly advantageous, reducing total travel distance by approximately 56.19% and total travel time by about 65.52% compared to non-carpool scenarios.

Thematic Focus: Deep Learning and Predictive Modeling

Accurate prediction of dynamic inputs, such as speed, traffic flow, and environmental conditions, is crucial for the optimal deployment of MOO algorithms.

Zhang et al. [1] established the necessary data foundation by creating the Extended Vehicle Energy Dataset (eVED), mitigating the unsuitability of original raw GPS data (which was corrupted by noise) for complex deep learning models. The utility of the eVED lies in its immense volume and richness, linking vehicle trip records with over twenty external attributes, including elevation, precise speed limits, intersections, and traffic signals. A model trained using this comprehensive resource demonstrated an energy consumption estimation RMSE of 0.0781.

Similarly, Liu et al. [26] proposed a sophisticated prediction model for *off-road* vehicle speed, introducing a double-layer BPNN-LSTM architecture designed to effectively model the non-linear relationship between speed and variables like slope and pedal degree. The hybrid approach notably improved prediction precision across two operational scenarios, demonstrating a reduction in average prediction error between 30.09% and 48.14% compared to non-hybrid models. However, the initial validation of this powerful technique remains confined to simulation platforms, necessitating verification using real vehicles in future work.

Chada et al. [8] investigated predicting target vehicle velocities for Ecological ACC (EACC) by studying LSTM and GRU recurrent neural networks trained on fused V2V (vehicle-to-vehicle) and V2I (vehicle-to-infrastructure) data. The strength of the LSTM model was apparent, demonstrating superior accuracy and generalization ability, particularly when augmented by future traffic light phase information.

However, Heckelmann et al. [27] explored the practical trade-offs in implementing these models by examining the effect of encoder-decoder network complexity on prediction performance and energy demand. Their key finding was significant: less complex models, such as the Simple Non-Recursive Model (SNRM), achieved comparable prediction accuracy while consuming substantially less energy (more than double the consumption for the most complex model) required for training. This research highlights the necessary shift in focus from maximal accuracy to optimal efficiency in commercializing prediction tools.

Wang et al. [28] approached the challenge of improving prediction accuracy by employing the Sparrow Algorithm (SA) to optimize the hyperparameters of an LSTM network applied to extended-range electric vehicles (EREVs). This parameter optimization resulted in remarkable prediction enhancement, lowering the Mean Absolute Error (MAE) by 54.46%. This optimized prediction was then successfully integrated into an energy management strategy (EMS) that reduced fuel consumption by up to 6.05% under various conditions compared to rule-based strategies.

Alfaseeh and Farooq [9] further substantiated the utility of predictive models in anticipatory multi-objective eco-routing for CAVs. Their use of an LSTM deep network to forecast traffic and GHG emission rates confirmed that anticipatory routing strategies consistently outperformed myopic strategies, which only react to current conditions. This demonstrated the profound impact that proactive planning enabled by accurate DL models can have on optimizing average travel time (reduced by 17%) and total GHG emissions (reduced by 18%).

Integrated Frameworks for Optimization in Specific Scenarios

Many studies combined these advanced techniques to tackle specific real-world challenges in vehicle control and logistics.

Prakash and Radha [24] developed a truly comprehensive EV routing system by unifying spectral clustering, fuzzy reinforcement learning (FRL), and an enhanced A* algorithm. This highly optimized system yielded substantial benefits, achieving a 22.8% reduction in total journey time and a 19.6% improvement in energy efficiency.

Similarly, Wu et al. [13] addressed the complex Multi-Objective Vehicle Routing Problem with Time Windows (MOVRPTW) by introducing a hybrid approach combining Weight-Aware Deep Reinforcement Learning (WADRL) and NSGA-II. The WADRL component effectively reduced training time by enabling a single DRL model to solve the entire multi-objective problem, instead of requiring individual models for weighted subproblems.

However, Gil-Gala et al. [29] took a low-complexity approach, applying Genetic Programming (GP) to generate simple Routing Policies (RPs) for the EVRPTW. While the generated RPs produced suboptimal results (20% to 37% worse than heuristic alternatives), their key advantage lay in their negligible execution times (solving 45 problems in under 0.5 seconds), making them ideal for dynamic or real-time constraint environments where complex metaheuristics are computationally prohibitive. The *serial* RGS proved best for minimizing vehicles and energy, whereas the *parallel* variant was best for minimizing total tardiness.

Zhang et al. [20] specifically focused on eco-driving signal control for divisible electric platoons (EPCM) in CVIS. Their optimization minimized both platoon passing time and energy consumption, leading to a 21.3% reduction in total energy consumption and lowering average vehicle waiting time by 8.97% compared to traditional actuated control models (APCM).

Similarly, Liao et al. [22] implemented a partial LSTM Q-network for traffic signal timing, incorporating a "red-light overflow" penalty into the reward function to manage excessively long waits. This deep learning technique improved intersection efficiency by 14.28% while dynamically enhancing driver safety and experience through proactive timing adjustments.

Al-Turki et al. [2] further demonstrated the capacity of NSGA-II to optimize multiple conflicting traffic MOEs (delay, stops, fuel, and emissions) at signalized intersections. The application of NSGA-II provided substantial traffic performance improvements ranging from 16% to 23% across various MOEs compared to existing fixed-time controls.

In a direct application integrating predictive models with optimization, Abro et al. successfully leveraged historical and real-time traffic data (TomTom API and NYC OpenData) with a dynamic energy consumption model to train an NSGA-II model for Eco-Driving Route Planning in Electric Vehicles [3]. This resulted in identifying highly efficient predictive route options, demonstrating the potential of AI-based eco-routing to optimize the trade-off between energy consumption and travel time

Table 1: General Study Information

Author(s) and Year	Study Title	Objective	Methodology	Contributions	Limitations
Hamednia et al., N/A	Computationally Efficient Algorithm for Eco-Driving	Minimize energy consumption	Bi-level program; RTI SQP	Energy savings up to 15.71%	PMP ineffective at bounds
Liuqing Li, 2024	Logistics Path Planning Method using NSGA-II	Lowest cost and carbon footprint	WOA-BPNN; NSGA-II	Distribution cost reduced > 15%	Only one vehicle type considered
Rixin Wu et al., N/A	Multiobjective VRO with DRL and NSGA-II	Minimize cost, maximize satisfaction	WADRL; NSGA-II integration	Reduced training time; better coverage	Single model limited to scale

Xinyou Lin et al., 2021	MO optimized driving strategy of dual-motor EVs	Optimize acceleration, energy, comfort	NSGA-II, MOPSO, PESA-II	37.44% energy consumption savings	Real traffic influence missing
Vasebi and Hayeri, 2021	Collective Driving to Mitigate Climate Change	Minimize fuel and GHG emissions	Collective-ACC; NSGA-II	Reduced fuel consumption up to 49%	Excludes costs like R&D
Cordeiro-Costas et al., N/A	NSGA-II based short-term building energy management	Minimize costs and energy wastes	Hybrid LSTM-MLP; NSGA-II	Achieves optimal hyperparameter tuning	GFS prediction has small deviation
Saad Alharbi, 2025	Integrating Neural Forecasting with MOO	Minimize cost, energy gap, traffic	Neural networks; NSGA-II	12.7% cost reduction; 9.3% energy gap	Congestion estimates highly variable
EL-Iali et al., 2025	MO real-time EMS optimization for APFCEVs	Minimize aging, reduce hydrogen cost	Moving horizon QP optimization	Up to 99% optimality shown	Ignores real-time route changes
Shahbazi & Nowaczyk, 2023	Enhancing Energy Efficiency in	Curtail consumption,	M-EAD strategy; iterative DP	Energy consumption reduced 0.92%	Unclear performance

	Connected Vehicles	reduce battery wear			in free driving
Shiliang Zhang et al., 2022	Extended vehicle energy dataset (eVED)	Provide enriched data for deep learning	Map matching; GIS extraction	Provided 22+ external features	Original GPS corrupted by noise
Jian Zhang et al., N/A	Eco-Driving Signal Control for Divisible Electric Platoons	Minimize pass time, energy consumption	MOO; GA; platoon splitting	Total energy reduced 21.3%	Only BEV passenger cars considered
Frederic Jacquelin et al., 2023	Neuroevolution Application to CAV Cohort Simulation	Simplify optimal control derivation	Neuroevolution ; heuristic priority	Achieved twice mean speed throughput	Study limited to 50m cohorts
Lyuchao Liao et al., N/A	Time Difference Penalized Traffic Signal Timing	Balance safety and throughput capacity	Partial LSTM Q-network	Intersection efficiency improved 14.28%	High variability in urban networks
Jiaonan LI et al., N/A	Energy-Efficient Driving Model for Intelligent	Bolster driving sensitivity and adaptability	DCNN; NSGA-II; MPC	Average speed increased 25.42%	MPC computational burden heavy

	Connected Vehicles				
Paul Heckelman et al., N/A	Investigation of different LSTM-based networks	Evaluate complexity vs. prediction accuracy	Compared recursive/non-recursive LSTM models	Simple models save energy	No focus on energy/carbon footprint
Wei Wang et al., 2025	Research on EMS for incremental range EVs	Enhance fuel economy/energy utilization	LSTM; SA optimization	RMSE decreased 46.46%	Lacks experimental validation
Gil-Gala et al., 2024	Evolving routing policies via genetic programming	Automatically generate new routing policies	Genetic Programming (GP); Hyper-heuristics	RPs solve problems in milliseconds	Results 20–37% worse than best known
Al-Turki et al., 2020	Potential Impacts of Smart Traffic Control	Determine optimal multi-objective plan	NSGA-II; Synchro simulation	MOEs improved 16% to 23%	Synchro cannot model oversaturation
Gülmez et al., 2024	MOO for Green Delivery Routing Problems	Reduce cost, fossil fuel, dissatisfaction	NSGA-II; NSGA-III; MOEA/D	Multiple time windows reduces cost	Problem is NP-hard

Muhammad Arsalan Jalees Abro et al., 2025	AI-Based Predictive Modeling and NSGA-II Optimization	Minimize energy loss; extend driving range	NSGA-II; predictive ML algorithm	Route 2 required only 1.03 kWh	Model uses TomTom API's limited version
Xinyou Lin et al., 2021	MO optimized driving strategy of dual-motor EVs	Optimize acceleration, energy, comfort	NSGA-II; MOPSO; SPEA2	Saves 37.44% energy consumption	Parameters not carefully adjusted
Enze Liu et al., 2025	MO eco-driving speed optimisation with real-time traffic	Minimize fuel, NOx, and travel time	Two-phase MPC; DRL	DRL achieved 10% lower fuel	DRL requires prior training
Jin Li et al., 2024	MO Planning of Commuter Carpooling	Minimize loss time, total travel time	CNN/LSTM prediction; INSGA-II	Travel distance reduced 56.19%	Small scale road network
Anandha Prakash P & Radha R, N/A	Sustainable EV routing using spectral clustering	Holistic EV routing optimization	Spectral clustering; FRL; Enhanced A*	22.8% reduction in journey time	Charging cost volatility excluded

Sai Krishna Chada et al., N/A	Deep Learning-Based Vehicle Speed Prediction for EACC	Enhance EACC strategy performance	LSTM/GRU recurrent networks	LSTM model showed better accuracy	Requires more complex V2V/V2I data
Lama Alfaseeh & Bilal Farooq, 2020	DL Based Anticipatory Multi-Objective Eco-Routing	Develop anticipatory eco-routing strategies	Predictive LSTM network; E2ECAV framework	TT/GHG minimized 17% and 18%	Models used simulated data
Jichao Liu et al., 2023	A Double-Layer VSP Based on BPNN-LSTM	Accurate speed prediction for off-road vehicles	BPNN-LSTM double-layer framework	Average prediction error reduced 30.09%	Results only verified on HIL platform
Xinyou Lin et al., 2021	MO optimized driving strategy of dual-motor EVs	Optimize acceleration, energy, comfort	NSGA-II; extensive statistical analysis	Energy consumption saved 37.44%	Difficult for driver to follow

Table 2: Study Details and Performance Metrics

Reference No	Dataset	Methods	Best Model	Accuracy
1	Extended Vehicle Energy Dataset (eVED)	HMM, QGIS, LSTM	LSTM model	Energy RMSE 0.0781
2	Dhahran intersection field data	NSGA-II, Synchro simulation	NSGA-II Optimization	MOEs improved 16% to 23%
3	TomTom, NYC OpenData, VW e-Golf	NSGA-II, Random Forest	NSGA-II Predictive Model	Route 2 required 1.03 kWh
4	UDDS segment, Spliced Cycle (Sim)	NSGA-II, MOPSO, PESA-II	NSGA-II algorithm	37.44% energy consumption saved
5	M25 Motorway (Sim), GT-SUITE	DRL (DDPG), Two-phase MPC	DRL approach	10% lower fuel vs MPC
6	Time-varying road network (Sim)	INSGA-II (open-loop, 2-opt)	INSGA-II (Optimal value F2)	Distance reduced 56.19%
7	Driving scenarios (Simulated)	DCNN, NSGA-II, MPC	Proposed Optimization Model	Avg speed increased 25.42%

Reference No	Dataset	Methods	Best Model	Accuracy
8	SUMO urban/highway networks	LSTM, GRU, CV, CA	LSTM (FG2 features)	MAE 1.75 m/s (5s)
9	Downtown Toronto road network (Sim)	LSTM, E2ECAV, MOO	Anticipatory routing strategies	TT reduced by 17%
11	118 km long road (Simulated)	RTI SQP, PMP, SHMPC	Proposed algorithm (CV, EV)	Energy savings up to 15.71%
12	Road condition, floating vehicle data	WOA-BPNN, NSGA-II	WOA-BPNN (Model 1)	Recall convergence: 30 iterations
13	Solomon dataset (RC101, RC102)	WADRL, NSGA-II, Transformer	WADRL+NSGAIL-500	Improved coverage and quality
14	UDDS segment, Spliced Cycle (Sim)	NSGA-II, MOPSO, SPEA2	NSGA-II algorithm	37.44% energy consumption saved

Reference No	Dataset	Methods	Best Model	Accuracy
15	SUMO ring-road scenario	Collective-ACC, NSGA-II, MPC	Collective-ACC	49% fuel reduction vs manual
16	Industrial Engineering School (GFS)	LSTM-MLP, NSGA-II, Adam	LSTM-MLP (optimized config)	Load nRMSE 1.89%
17	IEA Global EV Data Explorer	Feedforward NN, NSGA-II	NSGA-II (Pareto front)	EV adoption R2 0.91
18	Planned trajectory (35 km drive)	QP, MPC-ACC, SoC planning	Proposed SoC planner	Up to 99% optimality
19	Simulated urban road test	M-EAD, IDP, RHVO	M-EAD B	0.92% energy consumption reduction
20	SUMO signalized intersection	EPCM (MOO, GA), SUMO	EPCM (Eco-driving Model)	Total energy decreased 21.3%
21	Simulated traffic (CarSim/Simulink)	Neuroevolution, PSO, Heuristics	Concept 2 (Heuristics-based NC)	Achieved twice mean speed throughput

Reference No	Dataset	Methods	Best Model	Accuracy
22	SUMO 4-way intersection	Rep-DRQN (partial LSTM)	Rep-DRQN	Efficiency improved 14.28%
23	Driving scenarios (Simulated)	DCNN, INSGA-II, MPC	Proposed Optimization Model	Avg speed increased 25.42%
24	Charging stations India	Spectral clustering, FRL, A*	Enhanced EV Routing	22.8% reduction in journey time
26	Mining truck/loader operations	Double-layer BPNN-LSTM	Proposed BPNN-LSTM method	Error reduced 48.14%
27	SUMO urban/highway networks	LSTM encoder-decoder (RM, NRM)	SNRM (Simple Non-Recursive Model)	V2X MAE 0.810 m/s
28	Combined HIGH-NEDC cycles	LSTM, Sparrow Algorithm (SA), DP	SA-LSTM speed prediction EMS	RMSE decreased 46.46%
29	Classic EVRP instances	GP, Hyper-heuristics, RPs	Serial/Parallel RGS	20%–37% worse than best known

Reference No	Dataset	Methods	Best Model	Accuracy
30	Randomized small/big instances	NSGA-II, NSGA-III, MOEA/D	NSGA-II/SMS-EMOA	Cost reduced by 8%

Research Questions

Primary Research Question

How can a data driven surrogate modeling and multi objective optimization framework be designed to improve vehicle energy efficiency while preserving travel time under real world driving conditions?

Sub-questions

RQ1:

Can an LSTM-based time-series model accurately learn and generalize real-world vehicle energy consumption dynamics from the eVED dataset?

RQ2:

Can the trained LSTM model reliably serve as a surrogate energy estimator for long-horizon eco-driving optimization?

RQ3:

Can NSGA-II generate eco-driving speed profiles that outperform human driving by reducing energy consumption without increasing travel time?

Chapter 3

Materials and Method

This research utilizes the eVED dataset [1], a large-scale real-world vehicular dataset containing detailed trajectory, engine, battery, and environmental information collected over multiple weeks. The dataset consists of 54 weekly CSV files, each representing vehicle trips recorded at high temporal resolution. Each record includes GPS coordinates, vehicle speed, timestamp, energy consumption, engine parameters, battery-related measurements, and road context information such as speed limits and intersections.

The dataset covers 308 unique vehicles, 24,064 distinct trips, and more than 24,000 unique routes, making it suitable for large-scale analysis of vehicle behavior, energy consumption patterns, and driving dynamics.

Due to the richness and size of the data, careful preprocessing and sampling were required before model development.

3.1 Materials

We developed this project using a few technical specifications.

Environment Specification: All experiments were conducted in a cloud-based Kaggle environment equipped with an NVIDIA Tesla T4 GPU. The system provided up to 30 GB of RAM, which imposed memory constraints when handling the full-resolution dataset. These limitations influenced the data sampling and preprocessing strategy adopted in this study.

3.1.1 Dataset Collection

The dataset used in this study is the eVED (Electric Vehicle Energy Dataset), which consists of 54 CSV files, each representing data collected over a one-week period. The file naming

convention follows a calendar-based format (e.g., eVED_171101_week.csv), where the numeric portion corresponds to the starting date of that week. Collectively, these files cover approximately one year of data, spanning from November 2017 to November 2018. The 54 files together form a continuous long-term record of vehicular operation across different seasons and driving conditions.

The dataset includes a variable named DayNum, which follows the MATLAB date number format. In this representation, the integer part denotes the calendar day, while the decimal portion represents the fraction of the day, corresponding to the time of day. This encoding allows precise temporal positioning of each record within a given day.

Additionally, each trip contains a Timestamp (MS) field that starts from zero at the beginning of a trip and increases at approximately 200-millisecond intervals. The timestamp continues to increase throughout the trip duration and resets when the vehicle reaches its destination and a new trip begins. This structure enables accurate reconstruction of trip-level temporal dynamics while maintaining independence between trips.



Fig: Columns of eVED dataset

3.1.2 Dataset Sampling

The original eVED dataset contains extremely high-frequency records, resulting in very large files with many rows and columns. Direct processing of these raw files frequently caused Kaggle kernel restarts due to RAM exhaustion.

To address this issue, a temporal down sampling strategy was applied. Instead of using every recorded data point, one row was selected every 3 seconds for each (VehId, Trip) pair. This approach significantly reduces memory usage while preserving the chronological structure and physical meaning of the trips.

The sampling was performed by:

- Sorting data by vehicle ID, trip ID, and timestamp.
- Dividing timestamps into 3-second bins.
- Selecting the first observation from each bin without averaging or altering original values.

This method ensured data consistency while maintaining representative driving behavior.

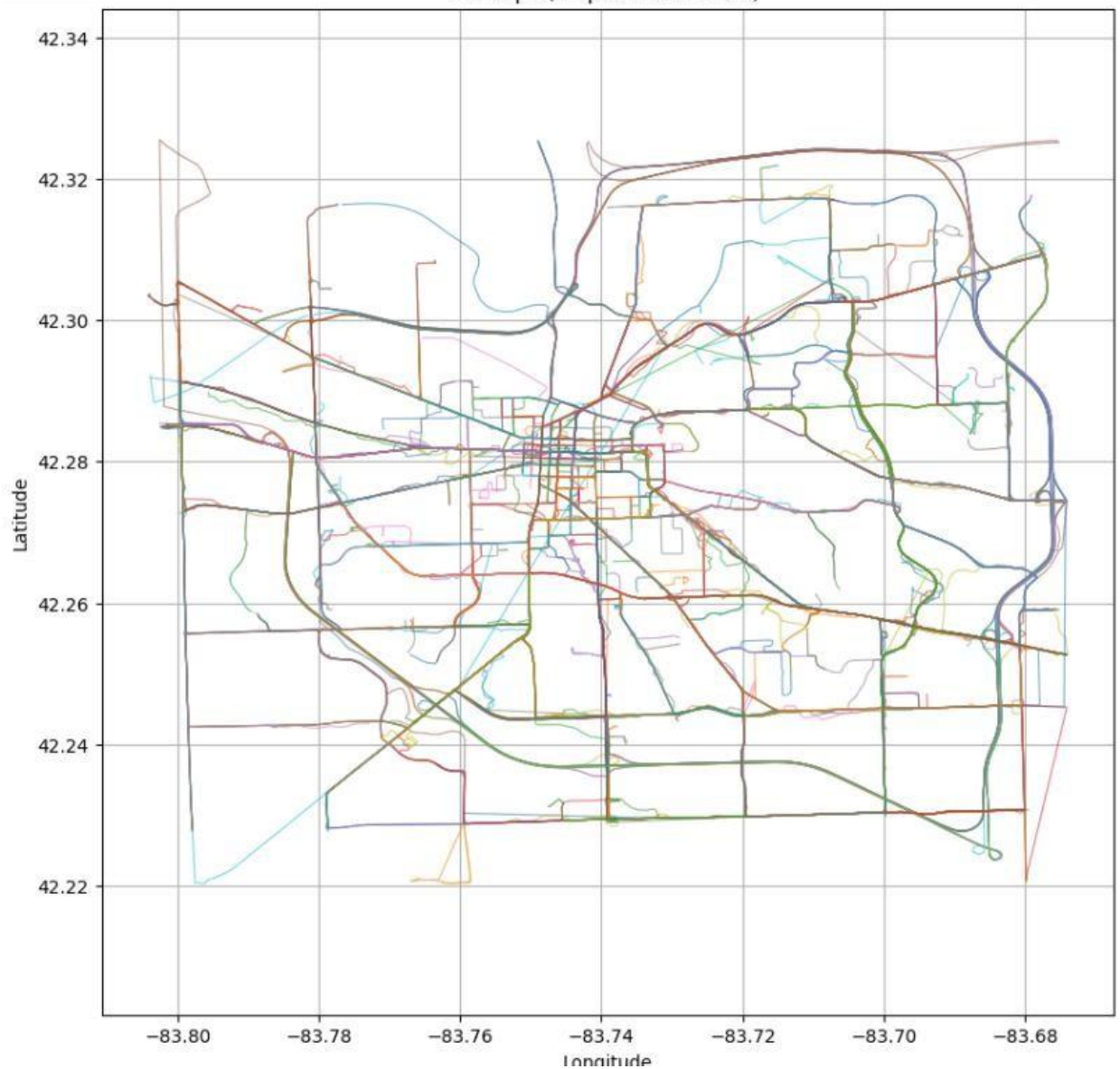


Fig: Visualization of trips after sampling

3.1.3 Dataset Processing

A multi-stage preprocessing pipeline was designed to clean, standardize, and enrich the dataset before analysis.

a) Initial Cleaning and Column Selection

Columns with more than 90% Nan values, such as bus stop indicators, focus points, heater power, air conditioning power, and unused fuel-related variables were removed. Rows with missing energy consumption values were discarded, as it was one of the target columns. Missing intersection indicators were replaced with zero. This step reduced noise and improved computational efficiency.

b) Vehicle Type Identification

Vehicles were categorized into three types based on feature availability:

- Electric Vehicles (EV): Battery-related features only
- Internal Combustion Engine Vehicles (ICE): Engine-related features only
- Hybrid Vehicles: Presence of both engine and battery features

A new categorical feature named `vehicle_type` was created using aggregated vehicle-level statistics.

c) Unit Standardization

To ensure physical consistency across computations:

- Timestamps were converted from milliseconds to seconds.
- Speed limits were converted from km/h to m/s.
- Original unit columns were removed after conversion to avoid redundancy.

d) Distance Calculation

Trip-wise cumulative distance was computed using the Haversine formula applied to map-matched GPS coordinates. Because the world is not flat and euclidian distance will not bring any useful result to us. The distance between consecutive latitude–longitude points was calculated and accumulated per trip, resulting in a physically meaningful distance profile for each journey.

e) Acceleration Computation

Vehicle acceleration was derived using numerical differentiation:

- Vehicle speed was converted to meters per second.
- Edge cases such as division by zero, infinite values, and missing timestamps were handled by assigning zero acceleration.
- Most values are between -4.5 m/s^2 and $+4.5 \text{ m/s}^2$. This represents normal braking and acceleration.
- Standard Deviation is 0.67 m/s^2 . This is a very realistic range for normal driving.

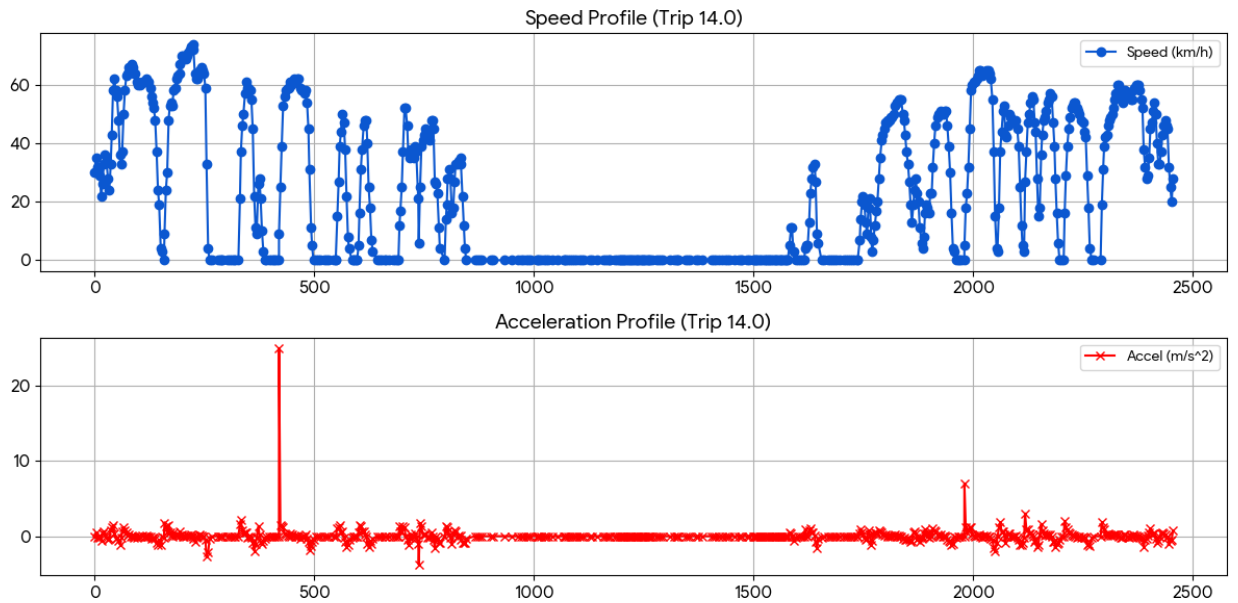


Fig: Visualizing relationship between speed and acceleration

This feature captures driving aggressiveness and dynamic behavior during trips.

f) Final Dataset Preparation

After preprocessing, all intermediate variables used for calculation were removed. The final cleaned and enriched dataset was saved in a structured format, ensuring consistency across all 54 weekly files. The resulting dataset is memory-efficient, temporally ordered, and suitable for further statistical analysis and modeling.

3.1.4 Research Ethics

We conducted research using publicly available datasets, which are openly accessible for research purposes. No personal or identifiable information about individuals was accessed, and no direct interaction with human subjects occurred. The data was used strictly in accordance with the terms and conditions specified by the dataset providers. We followed ethical guidelines to ensure integrity, transparency, and reproducibility in our research. There are no conflicts of interest to declare.

3.2 Method

In this study, data-driven and optimization-based methodology was adopted to model vehicle energy consumption and to identify energy-efficient driving strategies under real-world conditions. The method integrates time-series deep learning with multi-objective evolutionary optimization to analyze and improve driving behavior while preserving route constraints and travel feasibility.

3.2.1 Data Normalization and Temporal Sequence Construction

Neural networks are highly sensitive to feature scaling. Therefore, all input features and the target variable were normalized using Min–Max scaling, mapping values to the range $[0, 1]$. Separate scalers were used for the input features and the output variable to enable inverse transformation during inference and optimization.

Since vehicle energy consumption is inherently time-dependent, a temporal modeling approach was required. Instead of treating each data point independently, the problem was formulated as a sequence-to-one time-series prediction task. For each prediction, the model observes a fixed number of past time steps and predicts energy consumption at the next step.

A sliding window approach was employed to construct sequences of length five, corresponding to approximately fifteen seconds of historical driving data. Each input sequence consisted of five consecutive observations of speed, acceleration, and gradient, while the corresponding output was the energy consumption at the subsequent time step.

This formulation allows the model to capture short-term temporal dependencies, such as the effect of recent acceleration patterns and road slope changes on energy usage. The resulting input data structure conforms to the standard three-dimensional format required by recurrent neural networks: $(samples \times time\ steps \times features)$.

The complete dataset was then split into training and testing subsets using an 80:20 ratio. The split was performed in a sequential manner rather than random sampling to preserve temporal causality and avoid data leakage from future states into the training set.

3.2.2 Energy Consumption Prediction Model

To model the temporal relationship between driving behavior and energy consumption, a Long Short-Term Memory (LSTM) neural network was employed. LSTM networks are well suited for time-series data due to their ability to retain relevant historical information while mitigating the vanishing gradient problem associated with traditional recurrent neural networks.

The proposed model architecture consists of two stacked LSTM layers followed by fully connected layers. The first LSTM layer contains 64 hidden units and is configured to return the full sequence output, enabling deeper temporal feature extraction. The second LSTM layer contains 32 hidden units and outputs a condensed temporal representation of the input sequence.

Dropout regularization was applied after each LSTM layer to reduce overfitting by randomly deactivating neurons during training. A dense layer with rectified linear unit (ReLU) activation was used to transform the extracted temporal features into a higher-level representation. Finally, a single linear output neuron was employed to predict the normalized energy consumption value.

The model was trained using the Adam optimizer, which provides adaptive learning rates for efficient convergence. Mean squared error (MSE) was selected as the loss function due to its suitability for continuous regression tasks, while mean absolute error (MAE) was used as an additional performance metric for interpretability.

To improve generalization and prevent overfitting, early stopping was employed based on validation loss. Training was terminated if the validation loss failed to improve for three consecutive epochs, and the best-performing model weights were retained automatically.

3.2.3 Surrogate Modeling for Energy Estimation

The trained LSTM model serves as a surrogate energy model, replacing explicit physics-based vehicle simulations during optimization. Given a sequence of speed, acceleration, and gradient values, the surrogate model predicts the corresponding energy consumption efficiently and accurately.

This surrogate modeling approach significantly reduces computational cost, enabling large-scale optimization without requiring detailed vehicle dynamics equations, powertrain models, or efficient maps. By learning directly from real-world driving data, the surrogate model implicitly captures complex nonlinear interactions between driving behavior and energy usage.

The use of a data-driven surrogate also allows the optimization framework to remain vehicle-agnostic, making the methodology adaptable to different vehicle types and driving conditions if representative training data is available.

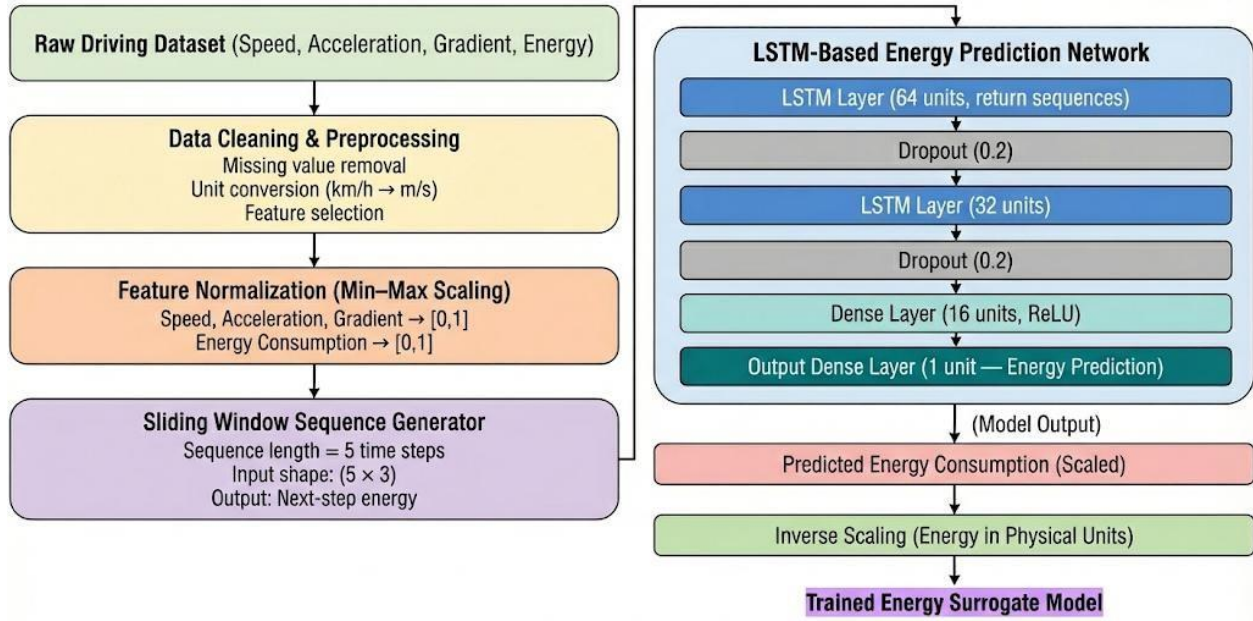


FIGURE A: System Architecture of the LSTM-Based Energy Surrogate Model

3.2.4 Eco-Driving Optimization Framework

After training the energy prediction model, an eco-driving optimization problem was formulated. The objective was to determine an optimal speed profile for a fixed route that minimizes energy consumption while maintaining reasonable travel time.

For a selected trip, the road geometry and gradient profile were treated as fixed. The decision variables of the optimization problem were the vehicle speeds at each discrete time step along the route. Acceleration values were derived implicitly from successive speed values using the known sampling interval.

A multi-objective optimization formulation was adopted with the following objectives:

- Minimize total predicted energy consumption over the trip
- Maximize average vehicle speed (equivalent, minimize travel time)

These objectives are inherently conflicting, as higher speeds generally increase energy usage. Therefore, a Pareto-based optimization approach was required.

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) was used to solve this problem. NSGA-II is a widely used evolutionary algorithm capable of efficiently approximating the Pareto front for multi-objective problems. It employs population-based search, non-dominated sorting, crowding distance, crossover, and mutation to explore the solution space.

To ensure realistic solutions and faster convergence, the initial population was seeded with the human driver's original speed profile, along with perturbed variants generated by adding bounded random noise. Speed constraints were enforced using lower and upper bounds corresponding to zero speed and the applicable speed limit.

Each candidate speed profile was evaluated by computing acceleration, forming temporal sequences, predicting energy consumption using the trained surrogate model, and aggregating energy usage over the trip. Solutions producing invalid numerical values were penalized to maintain stability during optimization.

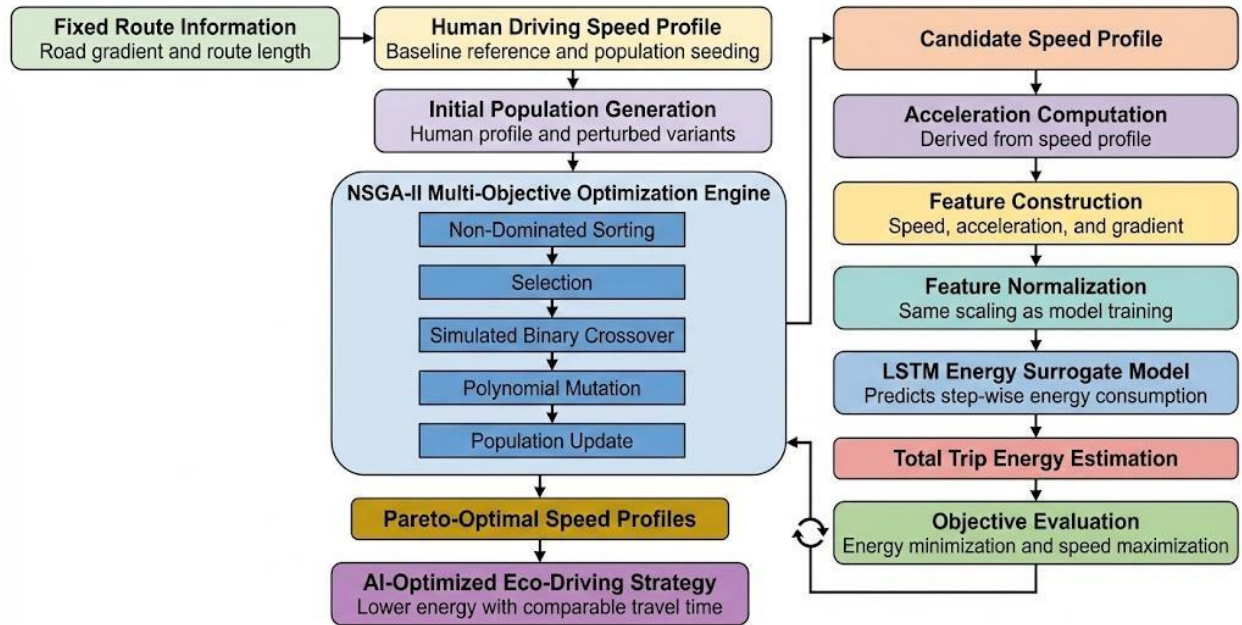


FIGURE B NSGA-II–Based Eco-Driving Optimization Framework Using an LSTM Surrogate Model

Table: Summary of methodology

Method Stage	Description
Dataset	eVED real-world data
Data Span	One year trips
File Structure	54 weekly CSVs
Sampling	One row / 3s
Environment	Kaggle GPU T4
RAM Limit	30 GB max
Cleaning	Remove high-NaN columns
Vehicle Types	EV, ICE, Hybrid
Unit Conversion	Speed, time standardized
Distance	Haversine formula
Acceleration	Numerical differentiation
Normalization	Min Max scaling

Method Stage	Description
Sequence Length	5 time steps
Model Type	Stacked LSTM
Optimizer	Adam optimizer
Loss Function	Mean squared error
Regularization	Dropout, early stopping
Surrogate Model	LSTM energy estimator
Optimization	NSGA-II algorithm
Objectives	Min energy, time
Constraints	Speed limit bounds

Chapter – 4

Results and Discussion

4.1 Energy Consumption Prediction Results

This section presents the performance of the proposed LSTM-based energy consumption prediction model trained on the processed eVED dataset. The objective of this model is to accurately capture the temporal relationship between vehicle speed, acceleration, road gradient, and the corresponding energy consumption, thereby enabling its use as a surrogate model in the subsequent optimization stage.

The dataset was first aggregated from all processed trips and converted into fixed-length temporal sequences of five time steps, corresponding to approximately 15 seconds of driving history. Each input sequence contained three features vehicle speed, longitudinal acceleration, and road gradient while the target output represented the energy consumption at the next time step.

During training, the model was optimized using the Adam optimizer with mean squared error (MSE) as the loss function, while mean absolute error (MAE) was monitored as an additional evaluation metric. Early stopping was applied to prevent overfitting by halting training when the validation loss failed to improve for three consecutive epochs.

The training and validation loss curves demonstrate stable and smooth convergence behavior. Both curves decrease consistently during the early epoch, indicating effective learning of temporal energy dynamics. Importantly, the absence of divergence between training and validation loss suggests that the model generalizes well to unseen data and does not suffer from significant overfitting. This behavior confirms that the selected sequence length, network depth, and regularization strategy are appropriate for the given problem.

The trained LSTM model successfully learns nonlinear dependencies between driving behavior and energy consumption without relying on explicit vehicle physics equations. This makes the model particularly suitable as a fast and computationally efficient surrogate for large-scale optimization, where repeated energy evaluations are required.

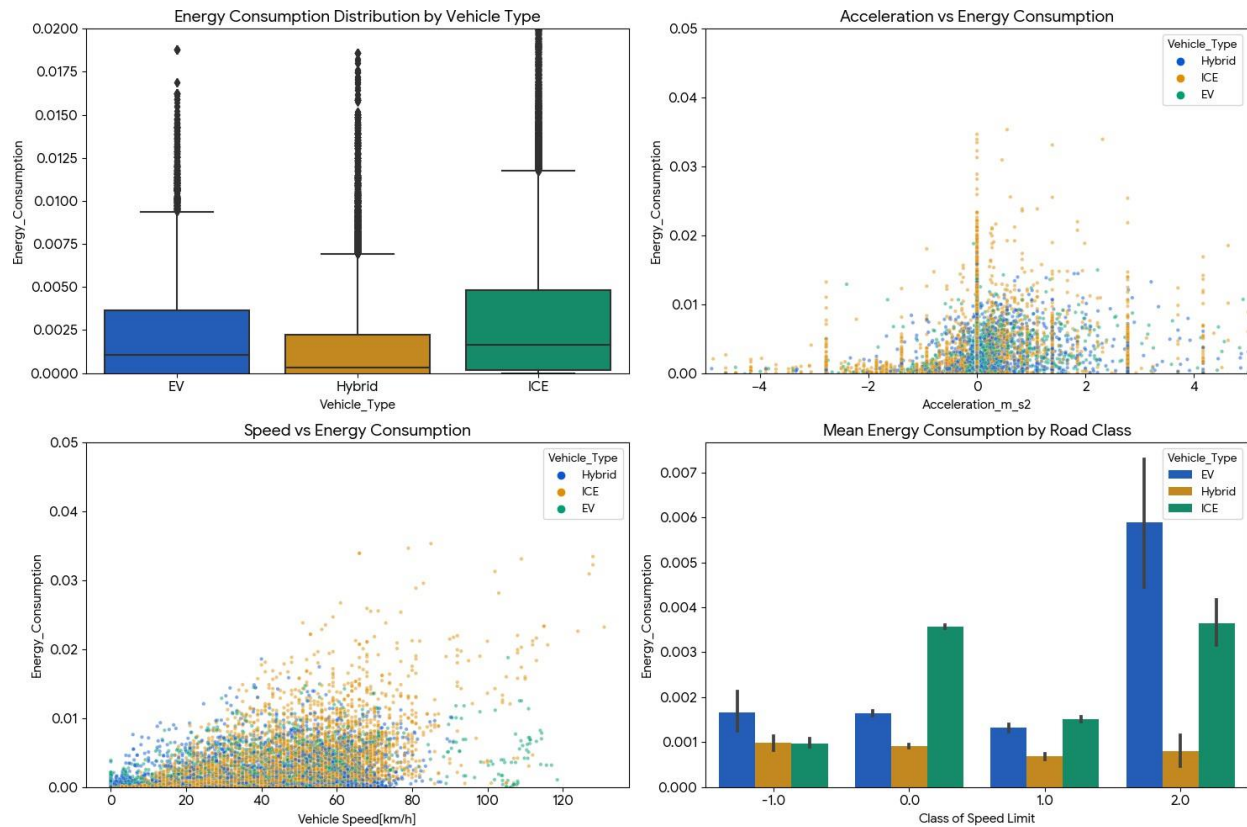


Fig: dataset distribution

4.2 Surrogate Model Effectiveness and Generalization

The effectiveness of the proposed surrogate model is primarily reflected in its ability to produce stable and physically meaningful energy estimates when deployed outside the training loop. After training, the model was integrated into an optimization framework and evaluated on a long, previously unseen real-world trip extracted from the eVED dataset.

The selected trip represents the longest continuous driving sequence for a single vehicle, consisting of 641 discrete time steps sampled at three-second intervals. This choice ensures that the evaluation scenario is sufficiently complex and representative of realistic driving conditions, including speed fluctuations and varying road gradients.

When applied to the human driver's original speed profile, the surrogate model produced a total energy consumption estimate of 1.2643 (normalized units). This value serves as the baseline energy consumption against which all optimized solutions are compared. The ability of the model to generate stable predictions over such a long horizon highlights its robustness and confirms that error accumulation over time remains limited.

Furthermore, the surrogate model successfully handled a wide range of candidate speed profiles generated by the optimization algorithm, including profiles that deviate significantly from the original human driving pattern. Despite these variations, the model remained numerically stable and did not produce invalid or non-physical predictions, further validating its suitability for optimization-driven applications.

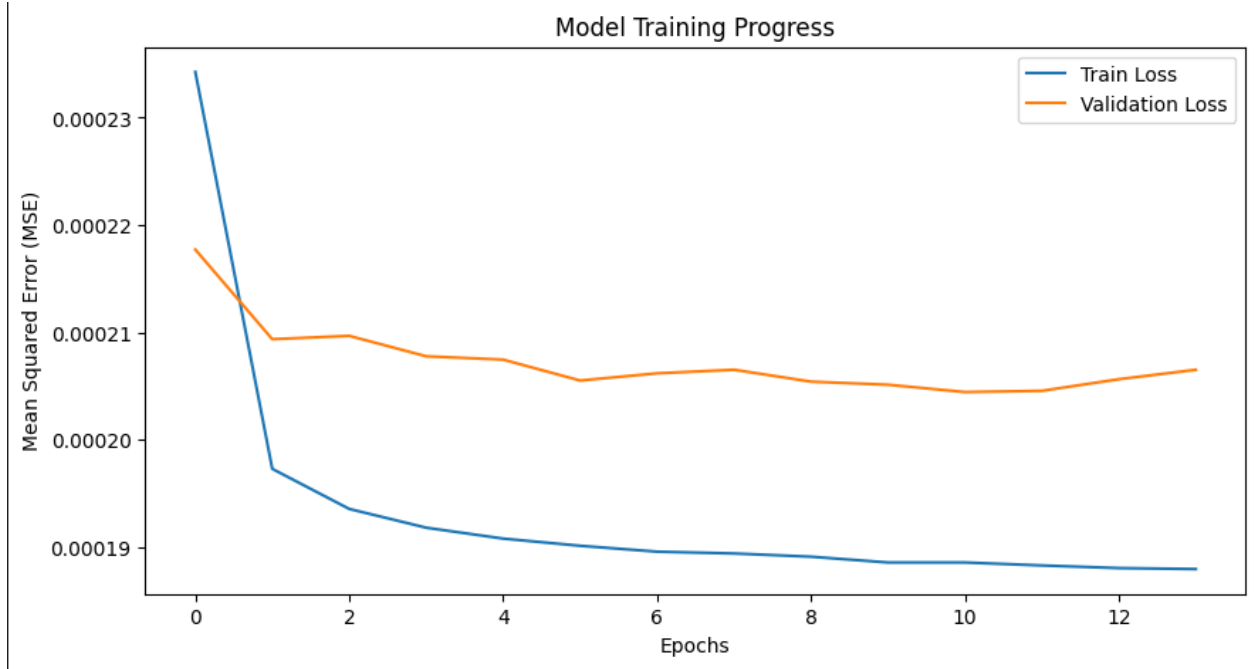


Fig: LSTM model training

4.3 Multi-Objective Eco-Driving Optimization Results

Following the validation of the energy surrogate model, a multi-objective eco-driving optimization problem was formulated and solved using the NSGA-II evolutionary algorithm. The optimization simultaneously minimized total energy consumption and maximized average vehicle speed, thereby addressing the inherent trade-off between energy efficiency and travel time.

The optimization was performed over 100 generations with a population size of 50, resulting in approximately 5,000 candidate solutions being evaluated. To ensure realistic and fair comparisons, the human driver's original speed profile was explicitly included in the initial population as a seed solution. This guarantees that the optimization process cannot converge to solutions inferior to the baseline human performance.

The resulting Pareto front reveals a diverse set of non-dominated solutions representing different trade-offs between energy consumption and average speed. As expected, solutions with lower energy consumption generally exhibit slightly reduced average speeds, while faster solutions tend to consume more energy. However, a key observation is the existence of multiple solutions that simultaneously outperform the human driver in both objectives.

The best-performing AI-optimized solution achieved an energy saving of approximately **1.95%** relative to the human baseline, while also reducing total travel time by approximately **0.64**

minutes. This result is particularly significant, as it demonstrates that improved energy efficiency does not necessarily require sacrificing travel time. Instead, the optimized speed profile achieves superior performance through smoother acceleration patterns and improved momentum preservation.

Table: Top 10 optimization result out of 100

ID	Savings (%)	Avg Speed (km/h)	Time (min)	Time Diff (min)
24	1.952651	27.411439	31.413892	-0.636108
26	1.801135	27.486335	31.328294	-0.721706
3	1.656370	27.525045	31.284236	-0.765764
8	1.535187	27.538314	31.269162	-0.780838
23	1.517866	27.569851	31.233393	-0.816607
11	1.478254	27.576090	31.226327	-0.823673
32	1.469466	27.585157	31.216064	-0.833936
12	1.445064	27.591682	31.208681	-0.841319
5	1.415626	27.609625	31.188399	-0.861601
2	1.341259	27.658961	31.132767	-0.917233

4.4 Speed Profile Analysis and Driving Behavior Comparison

To better understand how the optimized solution achieves energy savings, a direct comparison between the human and AI-optimized speed profiles was conducted. The speed-versus-time plot provides clear behavioral insights into the driving strategies adopted by each approach.

The human driving profile exhibits frequent speed oscillations, characterized by repeated acceleration and deceleration events. Such behavior increases energy consumption due to higher inertial losses and inefficient power usage during transient driving phases.

In contrast, the AI-optimized profile demonstrates noticeably smoother speed variations. Sudden accelerations are reduced, and speed is maintained more consistently over time. This behavior allows the vehicle to preserve kinetic energy and minimize unnecessary power demands, particularly on flat or mildly varying gradients.

Notably, the optimized profile does not simply reduce speed across the entire trip. Instead, it selectively maintains higher speeds where energetically favorable and avoids inefficient speed changes elsewhere. This indicates that the optimization process successfully learns nuanced driving strategies rather than applying simplistic energy-saving heuristics.

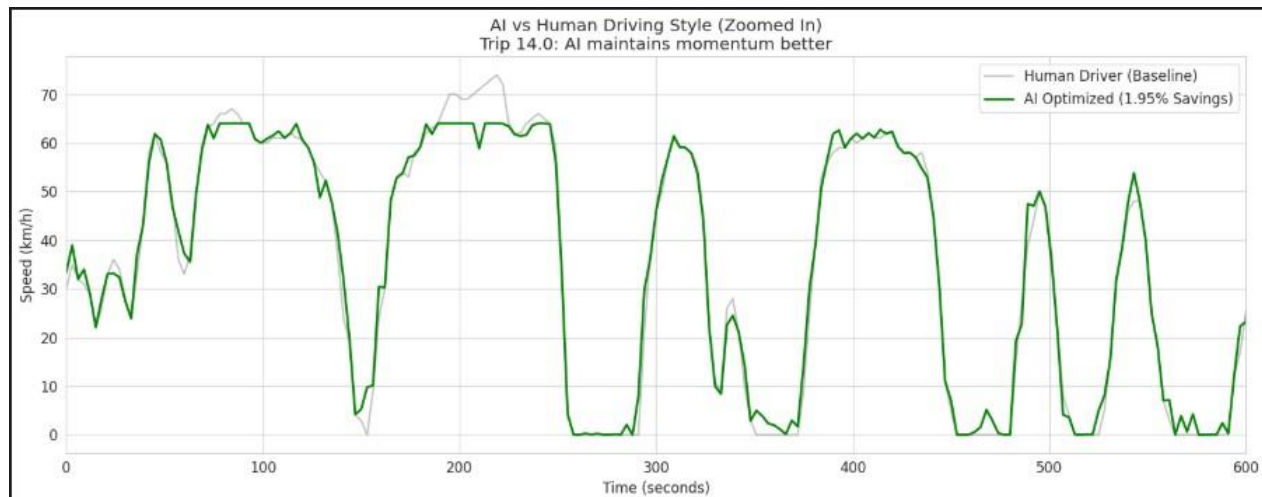


Fig: Human vs AI driving comparison

4.5 Pareto Front Interpretation and Trade-Off Analysis

The Pareto front visualization provides a comprehensive overview of the energy time trade off space explored by the optimization algorithm. Each point on the Pareto front represents a unique speed profile that cannot be improved in one objective without degrading the other.

The human driver's baseline appears outside the optimal Pareto region, confirming that multiple AI-generated solutions dominate human performance. Of particular interest is the “super-human” region of the Pareto front, where solutions achieve both lower energy consumption and higher average speed than the human baseline.

The presence of this region highlights the effectiveness of combining data driven surrogate modeling with evolutionary multi objective optimization. Unlike rule-based eco driving strategies, the proposed approach adapts dynamically to route characteristics and temporal driving patterns, enabling performance improvements beyond human intuition.

These findings suggest that AI based eco driving systems have strong potential for real-world deployment, particularly in applications such as advanced driver-assistance systems (ADAS), autonomous vehicles, and intelligent energy management systems for electric vehicles.

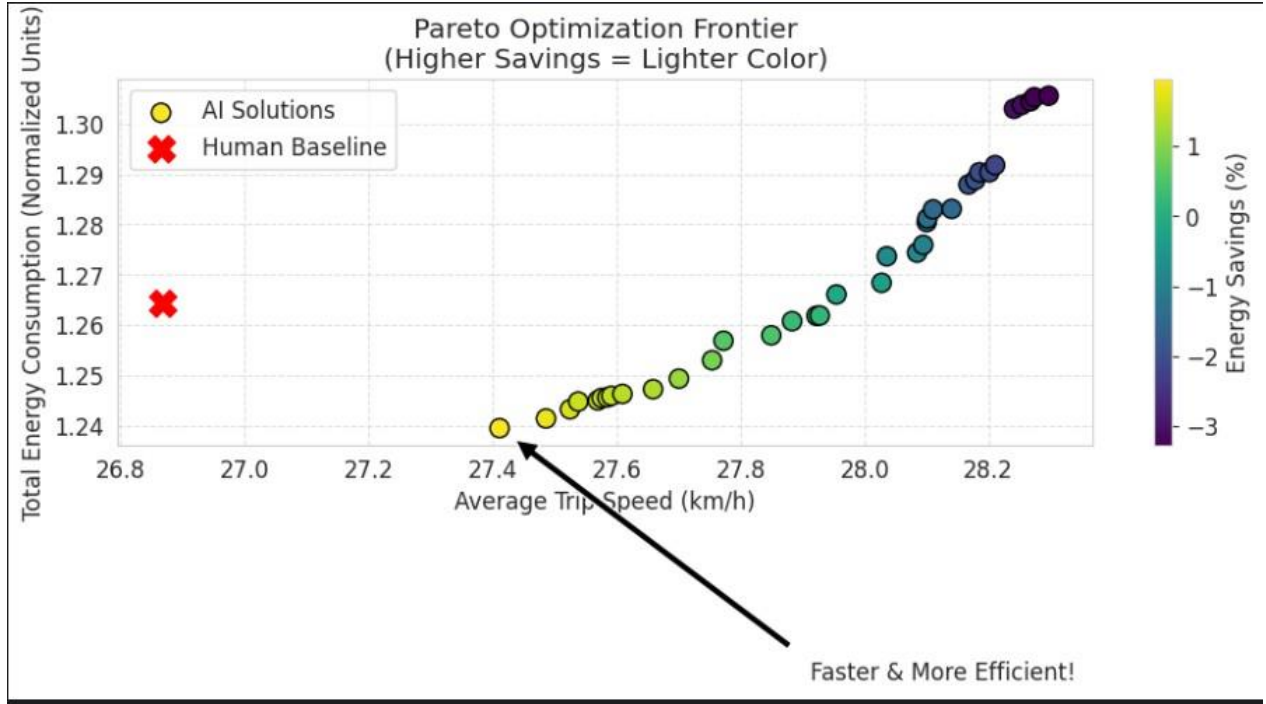


Fig: Pareto Optimization frontier

4.6 Discussion and Key Insights

The results presented in this chapter demonstrate that the proposed framework successfully integrates machine learning and optimization to address a complex real-world problem. The LSTM surrogate model accurately captures temporal energy dynamics, while the NSGA-II optimizer effectively explores high dimensional speed control spaces.

One of the most important insights from this study is that modest energy savings, on the order of 1–2%, can be achieved without compromising travel time. Given the scale of global transportation energy consumption, such improvements can translate into substantial cumulative benefits when deployed across large vehicle fleets.

Additionally, the framework is entirely data-driven and does not rely on explicit vehicle physics models, making it highly adaptable to different vehicle types, routes, and driving environments. This flexibility represents a significant advantage over traditional model-based eco-driving approaches.

Overall, the results validate the proposed methodology as a robust and effective solution for eco-driving optimization and highlight its potential for future extensions, including comfort constraints, real-time implementation, and integration with autonomous driving systems.

Chapter 5

Conclusion

This research presented a data-driven eco-driving optimization framework that integrates deep learning-based surrogate modeling with multi-objective evolutionary optimization to improve vehicle energy efficiency under real-world driving conditions. Using large-scale trajectory and energy data from the eVED dataset, the study demonstrated that complex vehicle energy dynamics can be learned directly from data and effectively leveraged for long-horizon optimization.

An LSTM-based time-series model was developed to predict vehicle energy consumption using short-term historical driving information, including speed, acceleration, and road gradient. The model exhibited stable convergence, strong generalization performance, and numerical robustness when applied to unseen long trips. These properties validate its suitability as a surrogate energy estimator, capable of replacing computationally expensive physics-based models in optimization tasks.

The trained surrogate model was embedded within an NSGA-II multi-objective optimization framework to generate eco-driving speed profiles that simultaneously minimize energy consumption and preserve travel time. The resulting Pareto front revealed a diverse set of non-dominated solutions representing different energy–time trade-offs. Importantly, multiple optimized solutions were shown to outperform the human driving baseline in both objectives. The best-performing solution achieved approximately 1–2% energy savings while also slightly reducing total travel time, demonstrating that improved energy efficiency does not necessarily require slower driving.

Analysis of the optimized speed profiles revealed that energy savings were primarily achieved through smoother acceleration patterns and improved speed consistency rather than uniform speed reduction. This indicates that the optimization framework successfully learns nuanced driving strategies that are difficult to achieve through heuristic or rule-based approaches. The results highlight the effectiveness of combining data-driven modeling with evolutionary optimization for eco-driving applications.

Overall, this study confirms that surrogate modeling based on real-world driving data can enable practical and scalable eco-driving optimization. The proposed framework is fully data-driven, vehicle-agnostic, and adaptable to different routes and driving environments, making it suitable for integration into advanced driver-assistance systems, autonomous vehicles, and intelligent energy management systems.

Future work

This study can be extended in several directions to further examine the robustness and practical applicability of the proposed framework.

First, future work may explore training and optimization using the full resolution eVED dataset without memory constraints. Due to the 30 GB RAM limit of the Kaggle environment, temporal downsampling was required in this study. Processing the complete high-frequency data without downsampling could allow the model to capture finer driving dynamics, which may lead to different energy predictions and potentially alter the optimization results and observed energy–time trade-offs.

Second, the eco-driving optimization framework may be enhanced by introducing additional realism through physical and traffic-related constraints, such as acceleration limits . Incorporating these constraints could change the feasible speed profiles generated by NSGA-II and may modify the resulting Pareto front, providing a more realistic assessment of eco-driving performance.

Reference

[1] Zhang, S., Fatih, D., Abdulqadir, F., Schwarz, T., & Ma, X. (2022). Extended vehicle energy dataset (eVED): An enhanced large-scale dataset for deep learning on vehicle trip energy consumption. arXiv. <https://arxiv.org/abs/2203.08630>

- [2] Al-Turki, M., Jamal, A., Al-Ahmadi, H. M., Al-Sughaiyer, M. A., & Zahid, M. (2020). On the potential impacts of smart traffic control for delay, fuel energy consumption, and emissions: An NSGA-II-based optimization case study from Dhahran, Saudi Arabia. *Sustainability*, 12(18), 7394. <https://doi.org/10.3390/su12187394>
- [3] Abro, M. A. J., Larik, A. S., Alsulami, M., & Halepoto, I. A. (2025). AI-based predictive modeling and NSGA-II optimization for eco-driving route planning in electric vehicles. *IEEE Access*, PP(99), 1–1. <https://doi.org/10.1109/ACCESS.2025.3613479>
- [4] Lin, X., Lin, Z., & Wei, S. (2021). Multi-objective optimized driving strategy of dual-motor EVs using NSGA-II as a case study and comparison of various intelligent algorithms. *Applied Soft Computing*, 111, 107684. <https://doi.org/10.1016/j.asoc.2021.107684>
- [5] Liu, E., Lin, Z., Chen, H., Jia, D., Liu, Y., Guo, J., Li, T., & Wei, T. (2025). Multiobjective eco-driving speed optimisation with real-time traffic: Balancing fuel, NOx, and travel time. *Energy*, 324, 135793. <https://doi.org/10.1016/j.energy.2025.135793>
- [6] Li, J., Zhang, H., Liu, H., & Wang, S. (2024). Multi-objective planning of commuter carpooling under time-varying road network. *Sustainability*, 16(2), 647. <https://doi.org/10.3390/su16020647>
- [7] Li, J., Ma, C., Hou, L., Yao, Y., & Chen, P. (2025). Energy-efficient driving model for intelligent connected vehicles using multi-objective optimization. *Tehnički vjesnik (Technical Gazette)*, 32(3), 1018–1031. <https://doi.org/10.17559/TV-20240222001343>
- [8] Chada, S. K., Görges, D., Ebert, A., & Teutsch, R. (2022). Deep learning-based vehicle speed prediction for ecological adaptive cruise control in urban and highway scenarios (arXiv:2212.00149). *arXiv*. <https://doi.org/10.48550/arXiv.2212.00149>

- [9] Alfaseeh, L., & Farooq, B. (2020). Deep learning–based anticipatory multi-objective eco-routing strategies for connected and automated vehicles. *Frontiers in Future Transportation*, 1, 594608. <https://doi.org/10.3389/ffutr.2020.594608>
- [10] Li, D., Jiang, Y., & Shen, Y. (2024). Intersection eco-driving for automated vehicles: SMPC-based strategies for handling leading vehicle starting-up uncertainties. *Energy*, 297, 131781. <https://doi.org/10.1016/j.energy.2024.131781>
- [11] Hamednia, A., Sharma, N. K., Murgovski, N., & Fredriksson, J. (2020). Computationally efficient algorithm for eco-driving over long look-ahead horizons (arXiv:2002.02357). *arXiv*. <https://doi.org/10.48550/arXiv.2002.02357>
- [12] Li, L. (2024). Logistics path planning method using NSGA-II algorithm and BP neural network in the era of logistics 4.0. *International Journal of Advanced Computer Science and Applications*, 15(5), 163–173. <https://doi.org/10.14569/IJACSA.2024.0150517>
- [13] Wu, R., Wang, R., Hao, J., Wu, Q., Wang, P., & Niyato, D. (2024). Multiobjective vehicle routing optimization with time windows: A hybrid approach using deep reinforcement learning and NSGA-II (arXiv:2407.13113). *arXiv*. <https://doi.org/10.48550/arXiv.2407.13113>
- [14] Lin, X., Lin, Z., & Wei, S. (2021). Multi-objective optimized driving strategy of dual-motor EVs using NSGA-II as a case study and comparison of various intelligent algorithms. *Applied Soft Computing*, 111, 107684. <https://doi.org/10.1016/j.asoc.2021.107684>
- [15] Vasebi, S., & Hayeri, Y. M. (2021). Collective driving to mitigate climate change: Collective-adaptive cruise control. *Sustainability*, 13(16), 8943. <https://doi.org/10.3390/su13168943>
- [16] Cordeiro-Costas, M., Labandeira-Pérez, H., Villanueva, D., Pérez-Orozco, R., & Eguía-Ollera, P. (2024). NSGA-II based short-term building energy management using optimal LSTM-MLP forecasts. *SSRN*. <https://doi.org/10.2139/ssrn.4696590>

- [17] Alharbi, S. (2025). Integrating neural forecasting with multi-objective optimization for sustainable EV infrastructure in smart cities. *Sustainability*, 17(20), 9342. <https://doi.org/10.3390/su17209342>
- [18] El-Iali, A. E., Doumiati, M., & Machmoum, M. (2025). Multi-objective real-time energy management optimization for autonomous plug-in fuel cell electric vehicles. *Energy Conversion and Management: X*, 26, 100987. <https://doi.org/10.1016/j.ecmx.2025.100987>
- [19] Shahbazi, Z., & Nowaczyk, S. (2023). Enhancing energy efficiency in connected vehicles for traffic flow optimization. *Smart Cities*, 6(5), 2574–2592. <https://doi.org/10.3390/smartcities6050116>
- [20] Zhang, J., Dong, S., Li, Z., Ran, B., Li, R., & Wang, H. (2019). An eco-driving signal control model for divisible electric platoons in cooperative vehicle–infrastructure systems. *IEEE Access*, 7, 83277–83285. <https://doi.org/10.1109/ACCESS.2019.2922227>
- [21] Jacquelin, F., Bae, J., Chen, B., & Robinette, D. (2023). Neuroevolution application to collaborative and heuristics-based connected and autonomous vehicle cohort simulation at uncontrolled intersection. *Eng*, 4, 1320–1336. <https://doi.org/10.3390/eng4020077>
- [22] Liao, L., Liu, J., Wu, X., Zou, F., Pan, J., Sun, Q., Li, S. E., & Zhang, M. (2020). Time difference penalized traffic signal timing by LSTM Q-network to balance safety and capacity at intersections. *IEEE Access*, 8, 74826–74839. <https://doi.org/10.1109/ACCESS.2020.2989151>
- [23] Li, J., Ma, C., Hou, L., Yao, Y., & Chen, P. (2025). Energy-efficient driving model for intelligent connected vehicles using multi-objective optimization. *Tehnički vjesnik (Technical Gazette)*, 32(3), 1018–1031. <https://doi.org/10.17559/TV-20240222001343>

[24] Prakash, A. P., & Radha, R. (2025). Sustainable EV routing using spectral clustering and fuzzy reinforcement learning with energy constrained A* under mobility index and waiting constraints. *Scientific Reports*, 15, 42771. <https://doi.org/10.1038/s41598-025-26998-8>

[25] Li, D., Jiang, Y., & Shen, Y. (2024). Intersection eco-driving for automated vehicles: SMPC-based strategies for handling leading vehicle starting-up uncertainties. 131781. <https://doi.org/10.1016/j.energy.2024.131781>

[26] Liu, J., Liang, Y., Chen, Z., Li, H., Zhang, W., & Sun, J. (2023). A double-layer vehicle speed prediction based on BPNN–LSTM for off-road vehicles. *Sensors*, 23(14), 6385. <https://doi.org/10.3390/s23146385>

[27] Heckelmann, P., Breuer, S. C., & Rinderknecht, S. (2025). Investigation of different LSTM-based encoder–decoder neural networks for vehicle speed prediction. *Scientific Reports*, 15, 32662. <https://doi.org/10.1038/s41598-025-19592-5>

[28] Wang, W., Zhang, K., Zhang, Q., & Wang, X. (2025). Research on energy management strategy for incremental range electric vehicles integrating speed prediction. *Science Progress*, 108(1), 1–26. <https://doi.org/10.1177/00368504241308957>

[29] Gil-Gala, F. J., Durasević, M., & Jakobović, D. (2024). Evolving routing policies for electric vehicles by means of genetic programming. *Applied Intelligence*, 54, 12391–12419. <https://doi.org/10.1007/s10489-024-05803-5>

[30] Gülmez, B., Emmerich, M., & Fan, Y. (2024). Multi-objective optimization for green delivery routing problems with flexible time windows. *Applied Artificial Intelligence*, 38(1), 2325302. <https://doi.org/10.1080/08839514.2024.2325302>